

Methodology: Implied Volatility Surface Completion

NQFO 2026 Olympiad Submission

Problem Statement

Given a partially observed implied volatility (IV) surface across four maturities (30, 60, 91, 182 days) and 15 moneyness levels ($K/S \in [0.80, 1.20]$) for both calls and puts, the task is to impute missing IV values while satisfying no-arbitrage constraints.

Phase 1 — Data Preparation

Input validation enforces positivity of spot prices and observed IVs. Two derived quantities are computed per row:

$$k = \ln\left(\frac{K}{S}\right), \quad w = \left(\frac{\sigma}{100}\right)^2 \tau,$$

where k is log-moneyness, w is total variance, and $\tau = T/252$ is time to expiry in years. These are the natural coordinates for no-arbitrage surface modelling.

Phase 2 — Volatility Regime Detection

Each trading date is mapped to one of three market regimes (*calm*, *normal*, *turbulent*) via k -means clustering ($k=3$) on five date-level features extracted from observed IVs: ATM IV level, term structure slope (30d→182d), put-call skew, smile curvature, and cross-strike IV standard deviation. Clusters are labelled by ascending ATM IV mean. Regime labels are used as features in the ML stage.

Phase 3 — Arbitrage-Free SVI Surface

An SVI (Stochastic Volatility Inspired) parametric smile is fitted independently per (**date**, **maturity**, **option_type**) slice using SLSQP minimisation of squared total-variance residuals:

$$w^{\text{SVI}}(k; \theta) = a + b\left(\rho(k - m) + \sqrt{(k - m)^2 + \sigma^2}\right).$$

The five parameters $\theta = (a, b, \rho, m, \sigma)$ are constrained to satisfy Lee's moment conditions (preventing arbitrage in the wings) and $g(k) \geq 0$ everywhere, where $g(k)$ is the Gatheral density function ensuring no butterfly arbitrage within the SVI surface itself. Multi-start optimisation (3 smart starts derived from ATM level and slope + 5 random starts) reduces sensitivity to local minima. Fitting uses observed IVs from both train and test dates (same-day known IVs are legitimate same-day anchors, not look-ahead). Calendar spread monotonicity $w(T_1) \leq w(T_2)$ for $T_1 < T_2$ is subsequently enforced by adjusting the level parameter a upward for the longer-maturity slice.

Phase 4 — ML Residual Correction

A `HistGradientBoostingRegressor` is trained on residuals $r = \sigma_{\text{obs}} - \sigma^{\text{SVI}}$ from training data only, using 16 features:

Feature group	Variables
Smile shape	$k, k^2, k^3, \tau, \ln \tau$
Type	<code>is_call</code>
Regime	regime label
Market context	ATM IV, term slope, skew, curvature, IV_σ
Interactions	$k\tau, k \cdot \text{regime}, k^2\tau$
SVI anchor	σ^{SVI}

Time integrity is preserved via expanding-window cross-validation (train on past, validate on strictly future dates). The final prediction is $\hat{\sigma} = \sigma^{\text{SVI}} + \hat{r}$.

Phase 5 — Arbitrage Enforcement

Calendar spread and **butterfly** violations are checked on predicted rows and corrected iteratively (up to 5 passes) by substituting the arbitrage-free σ^{SVI} value for any violating point.

Calendar: $w(k, T_1) \leq w(k, T_2)$ for $T_1 < T_2$ at fixed (k, type) .

Butterfly: $w(k_2) \leq \frac{(k_3 - k_2)w(k_1) + (k_2 - k_1)w(k_3)}{k_3 - k_1}$ for any adjacent triplet $k_1 < k_2 < k_3$.

Put-call parity: The robust mathematically sound arbitrage-free SVI surface ensures that put-call parity is followed in the SVI predicted values of IV, making the overall model financially realistic.

Overall, the math heavy SVI surface itself is sufficient to make most of the predicted IV values arbitrage free on its own, the further arbitrage enforcement and wing extrapolation correction in phase 5b perfectly handles the exceptional outliers, again showcasing robustness and financial correctness.

Phase 5b — Wing Extrapolation Correction

Diagnosis of large errors revealed that most of the high-error predictions were *wing extrapolations*: the target moneyness fell outside the observed strike range on that date. SVI parametric extrapolation is unreliable in this regime. The correction anchors each such prediction to same-day market data:

$$\hat{\sigma}_{\text{final}} = \alpha \hat{\sigma}_{\text{local}} + (1 - \alpha) \hat{\sigma}_{\text{SVI+ML}}, \quad \alpha = 0.9,$$

$$\hat{\sigma}_{\text{local}} = \sigma_{\text{boundary}}^{\text{obs}} + \Delta_{\text{train}}(m_{\text{boundary}} \rightarrow m_{\text{target}}),$$

where $\sigma_{\text{boundary}}^{\text{obs}}$ is the nearest observed IV on the same day, and Δ_{train} is the median IV difference between the two moneyness levels computed from training data.